Borrowing Strength Across Classes: Graph-Regularized Prototype Fields for Realistic Test-Time Adaptation of Vision-Language Models

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Abstract

Unlabeled test data can improve the zero-shot classifier of a vision-language model, but realistic evaluations show that transductive and test-time adaptation methods often fall below the zero-shot baseline when only a few vocabulary classes are present in a batch, when streams are correlated, or when the backbone changes. We trace a common cause to how adaptation is parameterised: every class owns independent parameters estimated from its own test samples, so absent or rare classes are either left at a biased text anchor or fitted to samples of other classes. We propose GRAFT (Graph-Regularized Anchor-displacement Field Transduction), which models the adapted prototypes as the text anchors plus a displacement field that is partially pooled over a semantic class graph built from the text encoder. A Gaussian Markov random field prior lets rare and absent classes borrow the displacement of their well-observed semantic neighbours, and an explicit class-presence model turns the vocabulary into a label-shift estimate instead of a fixed set of competitors. Inference is closed form and gradient free on cached features; the anchor-only shrinkage of prior work and a single global domain shift are the two extreme settings of one prior precision, whose strength can be chosen from the unlabeled batch by empirical Bayes. We show that absent classes receive the harmonic extension of the present-class displacements and that the posterior variance of every prototype is never larger than under independent shrinkage. The method is implemented with unit-tested closed-form updates and is evaluated under the realistic batch and online protocols on eleven benchmarks and ImageNet distribution shifts with several CLIP-style backbones; benchmark numbers are reported as the pre-registered experiments complete and no result in this draft is estimated.

1. Introduction

Vision-language models (VLMs) such as CLIP [24] and SigLIP [38] classify an image by comparing its embedding with text embeddings of class names. The zero-shot classifier is general but frozen: it ignores the structure of the data it is applied to. A growing family of methods therefore adapts it from the unlabeled test data itself, transductively on a batch [16, 19, 22, 36] or online on a stream [7, 18, 39, 41], training nothing and operating on cached features.

Two recent studies temper the enthusiasm. Zanella *et al*. [37] show that when a batch contains only a few of the vocabulary classes, which is the common case for a photo album, a surveillance feed or a user session, transductive methods lose tens of accuracy points with respect to the zero-shot model they are meant to improve, and Sheng *et al*. [29] report that test-time adaptation methods deliver marginal gains, transfer poorly to backbones other than CLIP and degrade calibration. The remedy proposed in [37], and refined in [14], is a statistical anchor: each class prototype is shrunk towards its text embedding with a strength that depends on how many test samples the class receives. Anchoring prevents collapse, but it treats every class as an island. A class that receives no samples stays exactly at its text anchor, although the anchor is known to be displaced from the image distribution by the modality gap and by the domain of the test data [20, 28]; a class with one or two samples is adapted from those samples alone.

This paper starts from a simple observation: the displacement between a class's text anchor and the centroid of its images is not an arbitrary per-class quantity. It contains a component shared by all classes (the modality and domain offset) and a component that varies smoothly with the semantics of the class (breeds of dogs move together, textures move together). We turn this observation into a prior. GRAFT models the adapted prototypes as $\mu_k = t_k + d_k$ and places a Gaussian Markov random field prior on the displacement field $D = [d_1; \ldots; d_K]$ whose precision is $\alpha I + \lambda L_T$, where L_T is the Laplacian of a k-nearest-neighbour graph over the text embeddings. The α term reproduces the anchor-only shrinkage of [14, 37]; the λ term is new and couples the classes, so that the displacement of a class with many samples propagates to its semantic neigh-

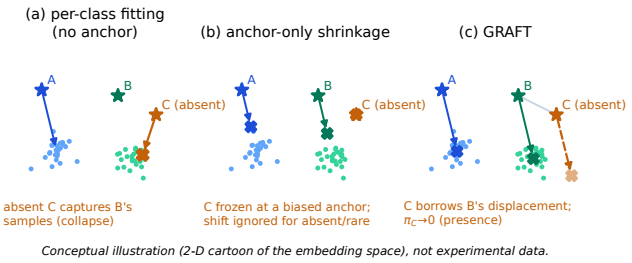


Figure 1. **Why realistic batches break per-class adaptation (conceptual).** Two classes (A, B) are present and class C is absent, while all image embeddings are shifted from their text anchors. (a) Fitting each prototype to its assigned samples lets the absent class C capture samples of B. (b) Anchor-only shrinkage leaves C at its biased anchor and ignores the shared shift for rare classes. (c) GRAFT pools the displacement field over the semantic class graph, so C inherits the displacement of its neighbour B, and the class-presence model removes C from the competition.

bours. The maximum a posteriori field has a closed form, $D = (N_z + s(\alpha I + \lambda L_T))^{-1} N_z E$, where N_z holds the soft counts and E the empirical displacements: a graph filter that returns the empirical displacement for well-observed classes and the harmonic extension of the neighbours' displacements for absent ones (Fig. 1c). Classes that receive no samples are additionally removed from the competition by a Dirichlet-regularised estimate of the class proportions, which treats the vocabulary as a label-shift problem [8, 27] rather than as a fixed set of equally likely competitors.

GRAFT is training free and gradient free: all updates are closed form on cached features, the online variant keeps a few sufficient statistics per class, and the prior strengths can be chosen by maximising the marginal likelihood of the unlabeled batch, removing the hyper-parameters that the critique of [29] targets. Our contributions are:

1. a formulation of transductive VLM adaptation in which the class prototypes form a displacement field with a graph-structured prior over the class vocabulary, together with the closed-form block-coordinate inference that fits it on batches and streams (Sec. 4);
2. an analysis showing that anchor-only shrinkage and a single global shift are the two extreme settings of the proposed prior, that absent classes receive the harmonic extension of present-class displacements through the Kron-reduced class graph, and that the posterior variance of every prototype is never larger than under independent shrinkage (Sec. 4.7);
3. an explicit class-presence model and an empirical Bayes rule for the prior strength, which remove the dataset-specific tuning of the pooling strength;
4. an implementation with unit-tested closed-form updates that plugs into the realistic batch and online protocols of [37], with a pre-registered experimental plan (Sec. 5).

2. Related Work

Transductive and test-time adaptation of VLMs. Transductive methods classify an unlabeled batch jointly. TransCLIP [36] fits a Gaussian mixture in the image embedding space, regularised by a KL term towards the zero-shot posterior and by a sample-graph Laplacian; EM-Dirichlet [22] works on the probability simplex; ZLaP [16] and ECALP [19] propagate labels over a nearest-neighbour graph of images and text prompts; LATA [1] and Conf-OT [31] use transduction to tighten conformal prediction sets. Online methods adapt over a stream: TDA [18] and DMN [41] keep caches of confident features, DPE [39] evolves text and image prototypes, BoostAdapter [40] bootstraps regional features, OnZeta [23] learns a label distribution and per-class vision proxies online, and OGA [7] maintains per-class Gaussians. Episodic methods act on single images through prompt tuning [30], augmentation ensembles [6, 35] or per-image prototype shifts [32]. GRAFT belongs to the first two families but changes the estimated quantity: a field over the vocabulary instead of a per-class parameter set.

Realistic evaluation and anchoring. StatA [37] introduces batches with a variable number of effective classes and correlated streams, documents the collapse of transductive methods in these regimes and proposes a statistical anchor that shrinks each class's Gaussian parameters towards the text-derived ones with strength $n_k/(n_k + \alpha)$. MOON [14] derives this shrinkage from penalised likelihood on the hypersphere with von Mises-Fisher components and makes its strength depend on zero-shot confidence at the instance and class levels. Both are per-class mechanisms: the anchor of class k is its own text embedding, and the information in other classes is not used. GRAFT keeps the anchor idea as the α part of its prior and adds the cross-class λ part; Sec. 4.7 shows that StatA's mean update is the exact $\lambda = 0$ special case of ours. Sheng *et al*. [29] benchmark TTA methods under one protocol and call for fewer tuned hyper-parameters and for backbone-agnostic designs, which motivates our empirical Bayes rule and our evaluation on SigLIP and OpenCLIP backbones.

Label shift and class presence. Adjusting a classifier to new class priors from unlabeled data is classical [8, 27]. For VLMs, Frolic [44] corrects the label bias of CLIP without labels, OnZeta [23] and LSA [10] estimate the target label distribution online, and the vocabulary scaling study of Chen *et al*. [2] shows that open-vocabulary classifiers degrade as distractor classes are added. Our presence model is a Dirichlet-regularised EM estimate of the proportions inside the same objective as the prototypes, so that an absent class loses both its prior mass and, through the field, its abil-

ity to act as a false attractor.

Structure over classes. Class graphs propagate classifier weights to unseen classes in zero-shot learning [17], propagate prototypes along a category hierarchy [21], and select correlated classes in CLIP test-time adaptation [13]; Laplacian regularisation over *samples* is standard in transductive few-shot learning [42, 45]. We are not aware of prior work that regularises the *displacement* of VLM prototypes from their text anchors with a Gaussian Markov random field over the class graph, nor of an analysis of what such a prior does for absent classes. The modality gap literature [20, 28] motivates the premise that a large part of the displacement is shared.

3. Preliminaries

Zero-shot classification. A VLM provides an image encoder f_v and a text encoder f_t mapping into a common space $\mathbb{R}^d$. For a vocabulary of K class names, prompt ensembling yields unit-norm text anchors $t_k = f_t(c_k)/\|f_t(c_k)\|$, collected in $T \in \mathbb{R}^{K\times d}$. A test image gives $x_i = f_v(I_i)/\|f_v(I_i)\|$, collected in $X \in \mathbb{R}^{N\times d}$. The zero-shot posterior is

$$\hat{y}_{ik} = \frac{\exp(\tau\, x_i^\top t_k)}{\sum_l \exp(\tau\, x_i^\top t_l)}, \tag{1}$$

with logit scale τ (100 for CLIP). We write Δ^K for the probability simplex and $\mathbf{1}$ for the all-ones vector.

Unlabeled adaptation. Given X and T only, the goal is to output posteriors $z_i \in \Delta^K$ that are more accurate and at least as well calibrated as $\hat{y}_i$. In the batch setting X is processed at once; in the online setting batches $X^{(1)}, X^{(2)}, \ldots$ arrive sequentially and each must be predicted when it arrives. Following [37], batches contain a variable number $K_{\text{eff}} \le K$ of effective classes and online streams are corre-

keeps $\mu_k = t_k$ even when every present class has moved by a common offset, (ii) a class with small n_k is estimated from its own few samples only, and (iii) the mixture weights are implicitly uniform, so absent classes keep competing for samples. GRAFT addresses (i) and (ii) by coupling the classes through a graph prior and (iii) by estimating the class proportions.

Semantic class graph. Let $S = TT^\top$ be the cosine similarity between class anchors. We connect each class to its m most similar classes with weight $w_{kl} = \exp((S_{kl} - 1)/\rho)$, symmetrise by taking the union of the two neighbourhoods, and form the normalised Laplacian $L_T = I - \mathrm{D}^{-1/2} W \mathrm{D}^{-1/2}$ with $\mathrm{D} = \mathrm{diag}(W\mathbf{1})$. L_T is positive semi-definite with spectrum in $[0, 2]$, is computed once per vocabulary from the text encoder only, and requires no image.

4. Method

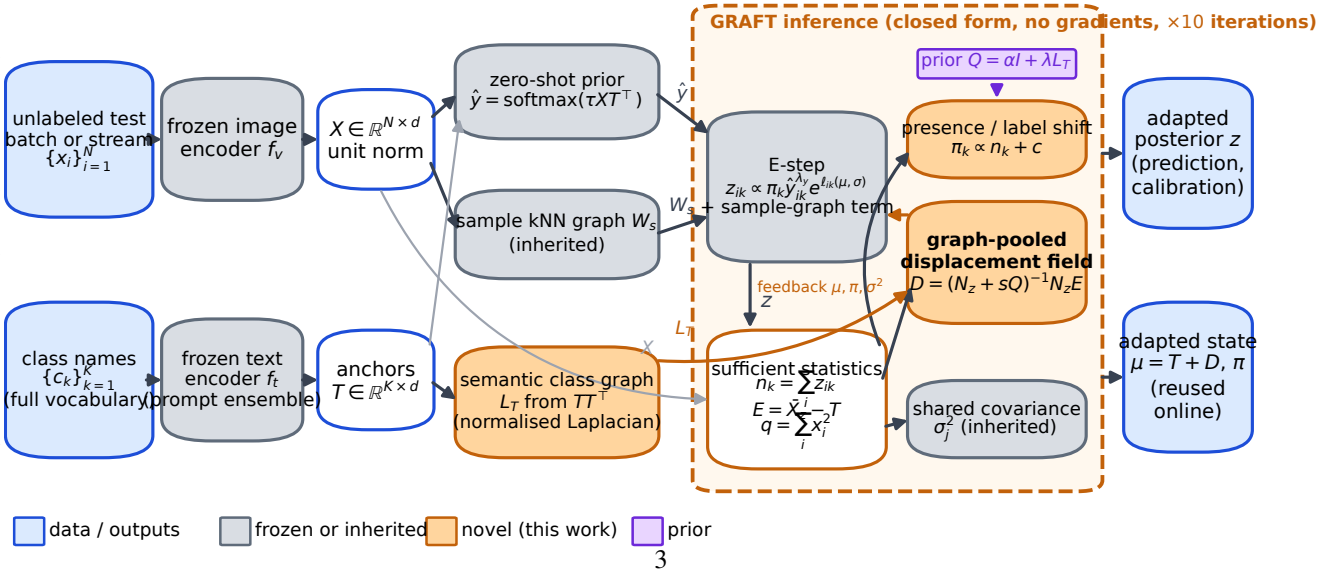


Figure 2. **Overview of GRAFT.** Frozen encoders produce unit-norm image features X and text anchors T (grey: frozen or inherited components; orange: this work). The text anchors define the zero-shot prior $\hat{y}$ and the semantic class graph L_T. Inference alternates a closed-form E-step over assignments z with closed-form M-steps for the class proportions π (presence), the graph-pooled displacement field D and the shared covariance. The adapted prototypes are $\mu = T + D$; the posterior z is the prediction. No gradient is computed and no parameter of the VLM is changed. In the online setting the sufficient statistics (n, E, q) persist across batches.

4.1. Framework overview

GRAFT keeps the generative view of transduction [36, 37] but changes the object that is adapted: instead of K independent prototypes it estimates a *displacement field* $D \in \mathbb{R}^{K\times d}$ anchored at the text embeddings, $\mu_k = t_k + d_k$, with a prior that couples semantically related classes, together with the class proportions $\pi \in \Delta^K$ and a covariance shared by all classes. Figure 2 shows the data flow: frozen encoders, the inherited sample graph W_s, the new class graph L_T, and a block-coordinate inference whose every step is closed form.

4.2. Graph-regularized displacement field

Likelihood. Each class is a Gaussian with mean μ_k and a diagonal covariance $\Sigma = \mathrm{diag}(\sigma^2)$ shared across classes. As in TransCLIP, the likelihood is tempered by κ, which plays the role of a covariance inflation and keeps the responsibilities soft on d-dimensional features:

$$\ell_{ik}(D,\sigma) = -\frac{1}{2\kappa}\Big[\sum_{j=1}^{d}\frac{(x_{ij}-t_{kj}-d_{kj})^2}{\sigma_j^2} + \sum_{j=1}^{d}\log\sigma_j^2\Big]. \tag{3}$$

Prior. The field is given a Gaussian Markov random field prior [26] that factorises over feature dimensions,

$$p(D) \propto \exp\Big(-\tfrac{1}{2}\sum_{j=1}^{d} d_{:j}^{\top} Q\, d_{:j}\Big), \qquad Q = \frac{\alpha I_K + \lambda L_T}{\kappa\,\bar{\sigma}^2}, \tag{4}$$

with $\bar{\sigma}^2 = \frac{1}{d}\sum_j \sigma_j^2$. The term αI_K pulls every displacement towards zero, that is, every prototype towards its text anchor; the term λL_T penalises the Dirichlet energy $\sum_{k\sim l} w_{kl}\|d_k/\sqrt{\mathrm{D}_k} - d_l/\sqrt{\mathrm{D}_l}\|^2$ and therefore asks the displacements of neighbouring classes to agree. Dividing by $\kappa\bar{\sigma}^2$ expresses α and λ in units of sample counts: a prior strength of $\alpha = 1$ is worth one tempered observation. Writing $N_z = \mathrm{diag}(n_1,\dots,n_K)$, $\bar{x}_k = \sum_i z_{ik}x_i/n_k$ and $e_k = \bar{x}_k - t_k$ (with $e_k = 0$ when $n_k = 0$), the maximum a posteriori field given the responsibilities is obtained for every feature dimension j from a $K \times K$ linear system,

$$d_{:j} = \big(N_z + s_j(\alpha I_K + \lambda L_T)\big)^{-1} N_z\, e_{:j}, \qquad s_j = \sigma_j^2/\bar{\sigma}^2. \tag{5}$$

Equation (5) is a graph filter applied to the empirical displacements, weighted by how much evidence each class has (Fig. 3). Classes with $n_k \gg \alpha + \lambda$ keep their empirical displacement; classes with little or no evidence receive a smoothed combination of their neighbours' displacements. The d systems share the matrix up to the scalar s_j and are solved jointly by preconditioned conjugate gradients, or exactly for small K.

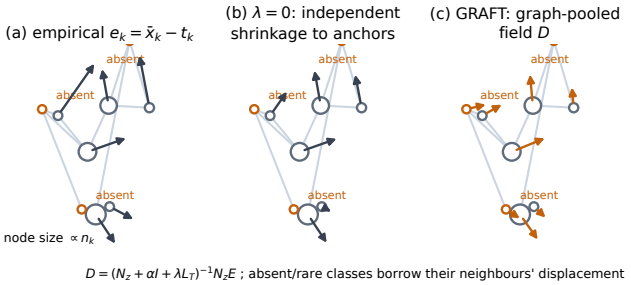


Figure 3. **The graph-pooled displacement field** on a toy class graph (node size: soft count n_k). (a) Empirical displacements are noisy for rare classes and undefined for absent ones. (b) Independent shrinkage ($\lambda = 0$) damps rare classes and leaves absent classes at their anchors. (c) With $\lambda > 0$, Eq. (5) transfers the displacement of well-observed classes along the graph; absent classes receive the harmonic extension of their neighbours (Proposition 2).

4.3. Class presence as label shift

The mixture weights π are free parameters with a symmetric Dirichlet prior $\mathrm{Dir}(c+1)$. Their M-step is the smoothed count

$$\pi_k = \frac{n_k + c}{N + Kc}, \tag{6}$$

and they enter the E-step as $\log\pi_k$, which is the classical label-shift correction of a calibrated classifier [27]: the zero-shot posterior implicitly assumes uniform class priors, and Eq. (6) replaces them by the estimate obtained from the batch. For a batch with $K_{\mathrm{eff}} \ll K$ classes, absent classes receive $\pi_k \approx c/(N+Kc)$ while present ones receive $n_k/(N+Kc)$; with $c = \frac{1}{2}$ a present class with sixteen samples is more than thirty times more likely a priori than an absent one in a batch of 64 over 1000 classes. The pseudo-count $c > 0$ keeps $\log\pi_k$ finite and lets a class that was absent in earlier online batches re-enter the stream.

4.4. Objective

Collecting the terms, GRAFT minimises over (z, D, σ, π)

$$J = \sum_{i,k} z_{ik}\big[\log z_{ik} - \log \pi_k - \lambda_y \log \hat{y}_{ik} - \ell_{ik}(D,\sigma)\big] - \frac{\lambda_L}{2m_s}\sum_{i,i'}(W_s)_{ii'}\, z_i^\top z_{i'} + \frac{1}{2}\sum_j d_{:j}^\top Q\, d_{:j} - c\sum_k \log \pi_k, \tag{7}$$

where the first line is the variational free energy of the mixture with the text prior $\hat{y}^{\lambda_y}$ of [36], the second line contains the inherited sample-graph term (m_s nearest neighbours), the new field prior and the presence prior. Setting $\lambda = 0$ and dropping the π terms recovers the StatA objective up to its per-class covariances; setting in addition $\alpha = 0$ and $\lambda_L = 0$ recovers the unanchored mixture.

4.5. Inference

Algorithm 1 alternates four closed-form blocks. The E-step is the standard mean-field update with the Laplacian term linearised at the current z (a majorise-minimise step [15, 36, 45]):

$$z_{ik} \propto \pi_k\, \hat{y}_{ik}^{\lambda_y} \exp\Big(\ell_{ik} + \frac{\lambda_L}{2m_s}\big[(W_s + W_s^\top)z\big]_{ik}\Big), \tag{8}$$

iterated a few times. The M-steps are Eq. (6), Eq. (5) and the shared variance $\sigma_j^2 = \frac{1}{N}\sum_{i,k} z_{ik}(x_{ij} - \mu_{kj})^2$. All quantities depend on the data only through the sufficient statistics $n = z^\top \mathbf{1}$, $S = z^\top X$ and $q = \sum_i x_i^{\odot 2}$, which makes the online variant immediate: statistics are accumulated over the stream with an optional forgetting factor ϱ, and the same M-steps are applied after every batch (algorithm in the supplementary material). The prediction for a batch is its posterior z after adaptation, as in the transductive protocol of [37]; a strictly inductive variant predicts before the statistics are updated.

4.6. Empirical Bayes selection of the pooling strength

Fixed (α, λ) are used for all datasets in the main protocol, but the model also admits a label-free rule. Given the responsibilities, the empirical displacements of the present classes $P = \{k : n_k \geq 1\}$ are, under the model, $e_P = d_P + \epsilon$ with $\epsilon_k \sim \mathcal{N}(0, \kappa\sigma^2/n_k)$ and $d_P \sim \mathcal{N}(0, \tilde{Q}_P^{-1})$, where $\tilde{Q}_P$ is the Schur complement of Q onto P. With the isotropic approximation $\sigma_j^2 \approx \bar{\sigma}^2$ the marginal likelihood of E_P is a Gaussian with covariance $C = \kappa\bar{\sigma}^2(\tilde{Q}_P^{-1} + N_{z,P}^{-1})$ per dimension,

$$\log p(E_P \mid \alpha, \lambda) = -\tfrac{1}{2}\operatorname{tr}\big(E_P^\top C^{-1} E_P\big) - \tfrac{d}{2}\log\det C + \text{const}, \tag{9}$$

which we maximise over a small logarithmic grid at every M-step whenever at least three classes are present (details in the supplementary material).

Algorithm 1 GRAFT, batch (transductive) inference

Require: features $X \in \mathbb{R}^{N\times d}$, anchors $T \in \mathbb{R}^{K\times d}$, graph L_T, (α, λ) or empirical Bayes, c, κ, λ_y, λ_L

$\hat{y} \leftarrow \mathrm{softmax}(\tau X T^\top)$; $z \leftarrow \hat{y}$; $D \leftarrow 0$; $\pi \leftarrow \mathbf{1}/K$; $\sigma^2 \leftarrow$ variance of X about T under $\hat{y}$

$W_s \leftarrow m_s$-NN cosine graph of X

for $t = 1, \ldots, T_{\max}$ **do**

  E-step: update z by Eq. (8) (n_{inner} sweeps)

  $n \leftarrow z^\top \mathbf{1}$; $\bar{X} \leftarrow N_z^{-1} z^\top X$; $E \leftarrow \bar{X} - T$ (rows with $n_k = 0$ set to 0)

  $\pi \leftarrow (n + c)/(N + Kc)$ ▷ presence

  optionally $(\alpha, \lambda) \leftarrow \arg\max$ marginal likelihood of E (Sec. 4.6)

  $D \leftarrow$ solution of Eq. (5) ▷ graph-pooled field

  $\sigma_j^2 \leftarrow \frac{1}{N}\big[q_j - 2\sum_k \mu_{kj} S_{kj} + \sum_k n_k \mu_{kj}^2\big]$ with $\mu = T + D$

end for

return z (posterior), (D, π, σ^2)

4.7. Analysis

All proofs are short and given in the supplementary material; the statements are also verified numerically by the unit tests shipped with the code.

Proposition 1 (Special cases)**.** *With $\lambda = 0$, Eq. (5) gives $d_k = \frac{n_k}{n_k + \alpha s_j} e_k$ per dimension, the count-based anchor shrinkage of StatA [37] (Eq. (2)) and MOON [14]. With $\alpha = 0$, L_T the Laplacian of the complete graph and $\lambda \to \infty$, every class receives the same displacement $d = \sum_k n_k e_k / \sum_k n_k$, the count-weighted mean displacement, that is, a single global shift of all prototypes.*

Proposition 2 (Absent classes are harmonic extensions)**.** *Let $A = \{k : n_k = 0\}$ and P its complement, and let $Q_\lambda = \alpha I + \lambda L_T$. The solution of Eq. (5) satisfies $Q_{\lambda,AA}\, D_A + Q_{\lambda,AP}\, D_P = 0$, so $D_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP} D_P$ is the harmonic extension of the present-class field, and D_P solves Eq. (5) restricted to P with Q_λ replaced by its Kron reduction $Q_\lambda / Q_{\lambda,AA}$ [5]. Adding absent classes to the vocabulary therefore changes the present-class prototypes only through the effective graph among them, never through data.*

Proposition 3 (Pooling never increases posterior variance)**.** *Under the model of Sec. 4.2, the posterior covariance of $d_{:j}$ given z is $\kappa\sigma_j^2 (N_z + s_j Q_\lambda)^{-1}$. Since $\lambda L_T \succeq 0$, for every class k, $\mathrm{Var}(d_{kj}) \leq \kappa\sigma_j^2/(n_k + s_j\alpha)$, the variance under independent shrinkage, with strict inequality for every class connected to a present class. For an absent class the bound under independent shrinkage is the prior variance $\kappa\sigma_j^2/(s_j\alpha)$, whereas with $\lambda > 0$ it is reduced by the neighbours' counts.*

Proposition 4 (Monotone block updates)**.** *Each block update of Algorithm 1 is the exact minimiser of Eq. (7) over its block when $\lambda_L = 0$, so J decreases monotonically and converges; with $\lambda_L > 0$ the E-step minimises a majoriser of J, as in [36].*

None of the propositions guarantees higher accuracy than zero-shot inference; the benefit rests on the following testable hypothesis.

Hypothesis 1 (Class-level smoothness)**.** *The oracle displacement field $e_k^\star = \mathbb{E}[x \mid y = k] - t_k$ has a Dirichlet energy on L_T that is substantially smaller than on a random graph with the same degree sequence. If this fails, λL_T cannot help and GRAFT reduces, in expectation, to the behaviour of anchor-only shrinkage.*

4.8. Complexity

Per outer iteration the E-step costs $O(NKd + Nm_sK)$, the statistics $O(NKd)$, the presence update $O(K)$ and the field $O(n_{\text{cg}}\, Kd\,(1{+}m))$ with conjugate gradients ($O(K^3{+}K^2d)$ exactly). Memory is $O(NK + Kd + K^2)$; the class graph has $O(Km)$ edges and is built once per vocabulary. Empirical Bayes adds $O(|G|K^3)$ per M-step for $|G|$ grid points, negligible for $K \leq 1000$ on a GPU. No backward pass is executed and the VLM is untouched after feature extraction.

5. Experimental Setup

All experiments operate on cached features; protocol, code and configurations are released. The benchmark runs are pre-registered but not yet executed; pending entries are marked TBD and no number in this manuscript is estimated.

5.1. Datasets

We follow the eleven-dataset suite of TransCLIP and StatA [36, 37] with the CoOp test splits [43]: ImageNet [4], SUN397, FGVC-Aircraft, EuroSAT, Stanford Cars, Food101, Oxford Pets, Flowers102, Caltech101, DTD and UCF101, plus the ImageNet distribution shifts ImageNet-V2 [25], ImageNet-Sketch [34], ImageNet-A [12] and ImageNet-R [11]. Class names and prompt templates are those of CoOp and StatA. Features are extracted once per backbone and cached; no training split is used.

The vocabulary sizes range from 10 (EuroSAT) to 1000 (ImageNet) and the test sets from 1,692 (DTD) to 50,000 images (ImageNet); the full list with split sizes is in the supplementary material.

5.2. Protocols

Batch adaptation with a variable number of effective classes. A task draws $K_{\text{eff}} \sim \mathcal{U}\{k_{\min}, \ldots, k_{\max}\}$ classes and a batch of B images from their union; the seven settings of [37] are $B{=}64$ with $K_{\text{eff}} \in [1,4]$, $[2,10]$, $[5,25]$ (very low, low, medium), $B{=}1000$ with $[5,25]$, $[25,50]$, $[50,100]$ (medium, high, very high), and the full test set, each averaged over 1000 tasks per dataset.

Online adaptation on correlated streams. Samples of every class are split over slots with Dirichlet(γ) proportions and the slots are concatenated, giving streams of batches of 128 with low ($\gamma{=}0.1$), medium (0.01) or high (0.001) correlation, plus the separate-class stream; 100 runs per dataset.

Full-set transduction and distribution shift. For comparison with TransCLIP-style evaluation we also run the full test set as one batch, and we evaluate the ImageNet shifts with the 1000-class (V2, Sketch) and 200-class (A, R) vocabularies.

5.3. Baselines

Published methods. Zero-shot CLIP; the episodic MTA [35]; the transductive EM-Dirichlet [22], ZLaP [16], ECALP [19], TransCLIP [36], StatA [37] and MOON [14]; the online TENT [33], TDA [18], DMN [41], DPE [39], BoostAdapter [40], OnZeta [23] and OGA [7], all with official code and default hyper-parameters through the StatA harness, into which GRAFT is plugged as a drop-in solver so that samplers, features and prompts are identical.

Architecture-matched controls. To attribute gains to the proposed mechanism rather than to the likelihood or the inherited terms, we run four controls inside our own implementation: anchor-only shrinkage ($\lambda{=}0$, the StatA/MOON mechanism in our likelihood), complete pooling (a single global shift; $\alpha{=}0$, complete graph, $\lambda \to \infty$), presence only (no displacement, π only) and the unanchored mixture ($\alpha{=}\lambda{=}0$, no π).

5.4. Implementation details

Inherited constants follow TransCLIP and StatA and are never tuned: $\kappa{=}50$, $\lambda_y{=}1$, $\lambda_L{=}1$, $m_s{=}3$, $T_{\max}{=}10$ outer iterations with 3 inner E-step sweeps. The new constants are fixed once for all datasets and backbones: $\alpha{=}1$ (one pseudo-count, as in StatA), $\lambda{=}4$, $m{=}5$ neighbours, $\rho{=}0.05$, $c{=}\frac{1}{2}$. The empirical Bayes variant searches $\alpha \in \{0.25, \ldots, 8\}$ and $\lambda \in \{0, 1, \ldots, 32\}$ on a logarithmic grid. The linear systems are solved exactly for $K \leq 256$ and by 30 preconditioned conjugate-gradient iterations otherwise. Backbones: CLIP ViT-B/16 (main), ViT-L/14 and RN50 [24], OpenCLIP ViT-B/16 trained on DataComp [3], and SigLIP ViT-B/16 [38], the backbone on which [29] found most TTA methods to fail. All runs use a single GPU; seeds and task indices are fixed and shared across methods.

5.5. Metrics

Top-1 accuracy averaged over tasks and then over datasets; the worst-decile accuracy (expected tail accuracy [7]) to expose collapses hidden by averages; expected calibration error with 15 bins [9] and negative log-likelihood of the adapted posterior, since [29] report that adaptation often trades calibration for accuracy; wall-clock time per task; and, for **??** 1, the ratio of the Dirichlet energy of the oracle displacement field on L_T to its energy on degree-preserving random relabellings of the classes.

6. Results

Entries marked TBD are pending; the released scripts regenerate every table from the result files.

6.1. Main comparison under realistic batches

Table 1 compares GRAFT with the published methods and the architecture-matched controls under the seven batch settings, averaged over the eleven datasets with CLIP ViT-B/16. The decisive comparison is between GRAFT and the anchor-only control, which differ only by the cross-class coupling λL_T and the presence model. The pre-registered criterion is that GRAFT is above zero-shot in every setting (no collapse) and above the anchor-only control in the very-low and low settings; a tie in the all-classes setting is expected since Proposition 1 makes the mechanisms coincide for large n_k.

6.2. Online adaptation on correlated streams

Table 2 reports the online protocol, where the failure mode of per-class methods is drift: in a correlated stream, prototypes of classes that have not appeared for a long time are stale or were fitted to foreign samples. GRAFT keeps absent classes tied to their neighbours through the field and lets the presence estimate follow the stream.

6.3. Backbones and distribution shifts

A table in the supplementary material tests the backbone-agnostic claim on OpenCLIP and SigLIP encoders and on the ImageNet shifts (low setting, B=64). Because GRAFT uses only cosine geometry, soft counts and the class graph, no component depends on CLIP-specific logit statistics beyond the inherited temperature.

6.4. Calibration and tail behaviour

A supplementary table reports expected calibration error, negative log-likelihood and worst-decile accuracy in the low setting. The presence model is expected to improve calibration on batches with few classes, because it removes probability mass from classes that are not there, whereas the field is expected to improve accuracy without changing the confidence structure; both expectations are testable in the table.

6.5. Testing the smoothness hypothesis

?? 1 is evaluated with labels, independently of any method: for every dataset we compute the oracle displacement field and the ratio of its Dirichlet energy on L_T to the mean energy over twenty degree-preserving random relabellings. A ratio well below one supports the hypothesis; a ratio near one on a dataset predicts that GRAFT matches but does not exceed the anchor-only control there, a dataset-level prediction that Tab. 1 can confirm or refute. The plot is produced by the released analysis script and will be added once computed.

7. Ablation Studies

All ablations use the low setting (B=64, $K_{\text{eff}} \in [2, 10]$) and the all-classes setting on the eleven datasets with CLIP ViT-B/16, so that each component is tested both where absent classes dominate and where they do not.

Components. Table 3 adds the components one at a time to the inherited mixture. The field prior and the presence model are the two novel components; the sample graph and the KL anchor are inherited and are also removed to measure their share.

Prior strength and graph. Table 4 varies α, λ and the number of neighbours m, and replaces the text kNN graph by a complete graph and by a degree-matched random graph. The random-graph row separates pooling from pooling *along semantics*: if it performs as well as L_T, **??** 1 is refuted for that benchmark.

Likelihood and optimisation choices. A supplementary table compares the shared diagonal covariance with an isotropic one and with the per-class covariances of StatA, re-projection of prototypes onto the sphere, the presence pseudo-count c, the inherited temperature κ, and the number of outer iterations, together with the measured time per task.

Failure cases. Two failure modes are anticipated and will be reported with examples: a class whose text neighbours are visually unrelated (a polysemous name) receives a wrong displacement when absent or rare, which the α term bounds and the random-graph control quantifies; and a batch mixing two domains, which a single field per class cannot represent (Sec. 8).

8. Conclusion

We traced the collapse of unlabeled VLM adaptation on realistic batches and streams to the independence of per-class parameters, which leaves absent and rare classes frozen at

Table 1. **Batch adaptation with a variable number of effective classes**, top-1 accuracy (%) averaged over 1000 tasks and the eleven datasets, CLIP ViT-B/16. Rows above the rule are published methods run with their official code through the common harness; rows below are architecture-matched controls sharing our likelihood. TBD: pending run. Best and second best will be marked in bold and underlined once the runs complete.

Method	Very low B=64	Low B=64	Medium B=64	Medium B=1000	High B=1000	Very high B=1000	All full
CLIP zero-shot [24]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MTA [35]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
EM-Dirichlet [22]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
ZLaP [16]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
ECALP [19]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
TransCLIP [36]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
StatA [37]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MOON [14]	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Unanchored mixture (α=λ=0, no π)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Presence only (π, no field)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Anchor-only shrinkage (λ=0)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Complete pooling (global shift)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GRAFT (fixed α, λ)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GRAFT + empirical Bayes	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 2. **Online adaptation**, top-1 accuracy (%) averaged over 100 runs and eleven datasets, batches of 128, CLIP ViT-B/16. Low/Medium/High: stream correlation $\gamma \in \{0.1, 0.01, 0.001\}$; Separate: classes arrive sequentially.

Method	Low	Medium	High	Separate
CLIP zero-shot	TBD	TBD	TBD	TBD
TENT [33]	TBD	TBD	TBD	TBD
TDA [18]	TBD	TBD	TBD	TBD
DMN [41]	TBD	TBD	TBD	TBD
OGA [7]	TBD	TBD	TBD	TBD
StatA [37]	TBD	TBD	TBD	TBD
MOON [14]	TBD	TBD	TBD	TBD
GRAFT (transductive within batch)	TBD	TBD	TBD	TBD
GRAFT (predict before update)	TBD	TBD	TBD	TBD

Table 3. **Component ablation**, top-1 accuracy (%) averaged over eleven datasets. π: presence model; α: anchor term; λ: graph coupling; EB: empirical Bayes; W_s: inherited sample graph.

π	α	λ	EB	W_s	Low (B=64)	All
zero-shot					TBD	TBD
–	–	–	–	✓	TBD	TBD
✓	–	–	–	✓	TBD	TBD
–	✓	–	–	✓	TBD	TBD
✓	✓	–	–	✓	TBD	TBD
–	✓	✓	–	✓	TBD	TBD
✓	✓	✓	–	–	TBD	TBD
✓	✓	✓	–	✓	TBD	TBD
✓	✓	✓	✓	✓	TBD	TBD

biased text anchors or fitted to foreign samples. GRAFT replaces independent prototypes by a displacement field over the class vocabulary with a Gaussian Markov random field prior on a semantic class graph, and replaces the implicit uniform class prior by an estimated one. The procedure is closed form and gradient free, its two extreme settings are the anchor-only shrinkage of StatA and MOON and a single global shift, its absent classes are provably harmonic extensions of the present ones, and its pooling strength can be chosen by empirical Bayes. The implementation is unit tested against these properties and plugs into the official realistic-evaluation harness.

Table 4. **Prior and graph ablation**, low setting, top-1 accuracy (%) averaged over eleven datasets. Default: α=1, λ=4, m=5, ρ=0.05.

α	0	0.25	1	4	16	
acc.	TBD	TBD	TBD	TBD	TBD	
λ	0	1	4	16	64	∞
acc.	TBD	TBD	TBD	TBD	TBD	TBD
graph	m=3	m=5	m=10	compl.	rand.	
acc.	TBD	TBD	TBD	TBD	TBD	

Limitations and future work. The benefit of the graph term rests on the class-level smoothness hypothesis, which we make explicit and test but which can fail for polysemous names; one field per class cannot represent a batch that mixes domains; the inherited tempered Gaussian likelihood is heuristic on the hypersphere and the empirical Bayes rule uses an isotropic approximation; and the benchmark evidence is pending (criteria in Sec. 6). Hierarchical class graphs and domain-mixture fields are natural extensions.

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LLM Usage Statement

In accordance with the CVPR 2027 call for papers, we disclose that a large language model based coding agent was used in the preparation of this submission: it assisted in surveying the literature, drafting the manuscript text, implementing the released code and producing the figure scripts, under the direction and review of the authors. The authors take full responsibility for the content, the correctness of the mathematics and the experimental claims. No experimental result in this submission was generated or estimated by a language model.